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REVIEW ARTICLE

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Breast cancer diagnosis: Imaging techniques and biochemical markers

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Breast cancer is a complex disease which is found as the second cause of cancer-associated death among women. Accumulating of evidence indicated that various factors (i.e., gentical and envirmental factors) could be associated with initiation and progression of breast cancer. Diagnosis of breast cancer patients in early stages is one of important aspects of breast cancer treatment. Among of various diagnosis platforms, imaging techniques are main diagnosis approaches which could provide valuable data on patients with breast cancer. It has been showed that various imaging techniques such as mammography, magnetic resonance imaging (MRI), positron-emission tomography (PET), Computed tomography (CT), and single-photon emission computed tomography (SPECT) could be used for diagnosis and monitoring patients with breast cancer in various stages. Beside, imaging techniques, utilization of biochemical biomarkers such as proteins, DNAs, mRNAs, and microRNAs could be employed as new diagnosis and therapeutic tools for patients with breast cancer. Here, we summarized various imaging techniques and biochemical biomarkers could be utilized as diagnosis of patients with breast cancer. Moreover, we highlighted microRNAs and exosomes as new diagnosis and therapeutic biomarkers for monitoring patients with breast cancer.

KEYWORDS
biomarker, Breast cancer, diagnosis, imaging technique, therapy

1 | INTRODUCTION

Breast cancer is found as one of major health public problems among various women which a variety of factors such as genetical and envirmental factors could affect on initiation and progression of it (Singletary, 2003). Early detection and monitoring of patients in response to various types of therapies are important aspects of breast cancer therapy (Barber, Jack, & Dixon, 2004). The utilization of imaging techniques is one of main approaches for detection and assessment response to therapy in patients with breast cancer (Khalkhali, Mena, & Diggles, 1994). Multiple lines of evidence indicated that various imaging techniques such as mammography, MRI, PET, CT, and SPECT could be employed as effective tools for assessment patients with various cancer such as breast cancer, giloma, oral cancer, and

melanoma (Keshavarzi, Darijani, et al., 2017; Keshavarzi, Sorayayi, et al., 2017; Khanmohammadi, Mir, Baniebrahimi, & Mirzaei, 2017; Saadatpour et al., 2017; Saadatpour, Bjorklund, et al., 2016). Growing evidences revealed that imaging techniques are associated with some limitations such as expensive (Andreea et al., 2011). Hence, it seems that utilization of improvement imaging techniques and biochemistry biomarkers could overcome to limitations associated with imaging techniques. Beside, utilization of imaging techniques, identification of new biomarkers in patients with breast cancer patients has provided an interesting landscape (Duffy et al., 2017). It has been showed that a variety of biochemistry biomarkers (i.e., measurement of enzymes, hormones, and expression profiles of mRNAs, microRNAs). Recently, MicroRNAs (miRNAs) and exosomes have been emerged as new diagnostic and therapeutic biomarkers for breast cancer patients (Santos, Ribeiro, Sarian, Ortega, & Derchain, 2016). These molecules play critical roles in breast cancer pathogenesis via targeting a variety of cellular and molecular pathways involved in breast cancer pathogenesis (Kurozumi et al., 2017). It has been showed that alteration of miRNAs expression (up/down regulation) could be affected on initiation and progression of various cancers such as breast cancer (Gholamin et al., 2017; Kurozumi et al., 2017; Mirzaei, Fathullahzadeh, et al., 2017; Moridikia, Mirzaei, Sahebkar, & Salimian, 2017). Here, we summarized various imaging techniques are used for detection and monitoring patients with breast cancer. Moreover, we highlighted utilization of various biomarkers such as measurement of enzymes, miRNAs, and exosomes as diagnostic and therapeutic biomarkers for patients with breast cancer.

2 | IMAGING TECHNIQUES AND BREAST CANCER

Breast cancer is one of important malignancies worldwide which has accounted as the most cancer among women. Early detection and monitoring of patients are found as main aspects of breast cancer therapy (Weigel & Dowsett, 2010). Breast cancer could be diagnosed by monitoring of patients with imaging techniques or symptoms such as pain or a palpable mass. It has been showed; the monitoring healthy subjects could contribute to detection of small tumors which could lead to progression of breast cancer (Weigel & Dowsett, 2010). Among of various techniques, imaging techniques have been emerged as powerful tools for detection and monitoring response to therapy in patients with breast cancer (Table 1) (Alcantara, Leal, Garcia-Bocanegra, & Garcia-Martin, 2014). Mammography is known as gold standard techniques for detection of breast cancer patients which are associated with various benefits such as high sensitivity and specificity, inexpensive, and well tolerated (Wellings, Vassiliades, & Abdalla, 2016). It has been showed that using of mammography could reduce 19% breast cancer mortality. Besides, various benefits of mammography, there are some limitations which are associated with using this technique such as pain and anxiety, false alarms, and radiation risks (Wellings et al., 2016). Hence, it seems that utilization of new techniques with better properties are needed.

Ultrasounds is other major imaging techniques which are known as other most common method for diagnosis and monitoring response to therapy in breast cancer (Ozmen et al., 2015). Besides, mammography, using of ultrasound is required for the management and diagnosis of breast pathology (Ozmen et al., 2015). The utilization of ultrasound are associated with various advantages (e.g., it does not use ionizing radiation, and high sensitivity), which have provided it as powerful diagnosis tools for detection of breast tumors in young women and pregnant and breastfeeding women (Alcantara et al., 2014). Ultrasound techniques such including Ultrasound contrast and Elastography are also applied for breast intervention, and for the placement of harpoons (Alcantara et al., 2014).

In a study, Jia et al. (2014) assessed breast tumor angiogenesis by three-dimensional contrast enhanced ultrasound (3D-CEUS) and dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) score systems as diagnosis platforms in 35 benign and 148 malignant breast tumors. They results indicated that 3D-CEUS score system could be used as an effective diagnosis plathform for evaluation of tumor angiogenesis in patients with breast cancer. Their results showed that this score system shows a well association with various tumor perfusion and vascular characteristics such as microvessel density (MVD), vascular endothelial growth factor (VEGF), and matrix metalloproteinases (MMP-2, MMP-9) expression (Jia et al., 2014).

MRI is found as other important diagnostic tools for breast cancer. This technique could be used for various aspects of breast cancer treatment including monitoring of response to therapy, monitoring high risk patients, assessment of breast cancer metastasis, and study of tumor recurrence (Mann, Kuhl, Kinkel, & Boetes, 2008). The provided data by MRI, could suggested various actions (e.g., breast discharge, staging of breast cancer, evaluation of microcalcifications, premalignant lesions, and residual tumor in operated patients) to clinicians (Sardanelli et al., 2010). These properties led to introduced MRI as a suitable diagnosis technique for patients with breast cancer (Mann et al., 2008; Sardanelli et al., 2010).

The mentioned techniques including mammography, ultrasounds, and MRI could be employed as effective tools for detection and monitoring patients with breast cancer (Alcantara et al., 2014). On the other hands, these techniques are associated with various limitations such as lacks of sensibility. Hence, radiologist has efforted to introduce new techniques which could overcome to the current limitations (Alcantara et al., 2014). It has been showed that utilization of external substances called contrast agents which are able to improve the sensibility of images in the region of interest, could be effective platforms for diagnosis and monitoring patients in early stages of various cancers such as breast cancer (Alcantara et al., 2014). The using of contrast agents could enhance the quality and the follow-up of molecular processes at the cellular and molecular levels of the region under assessment. The gadolinium- and iodine-based structures are known as common contrast agents which could be used for MRI and mammography (X-ray), respectively (Alcantara et al., 2014).

PET and SPECT are other imaging techniques which could be used for diagnosis breast cancer patients. PET technique uses radioactive isotopes which emit positrons (e.g., ^{18}F, ^{15}O, ^{13}N, and ^{11}C) whereas

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TABLE 1 The used imaging techniques for detection breast cancer patients

Techniques	Model	Samples	Sensitivity and specificity	Ref
PSOWNN	Human	54		(Dheeba, Singh, & Selvi, 2014)
Ultra-wideband				(Lai, Soh, Gunawan, & Low, 2011)
MR imaging	Human	70	Sensitivity 97%	(Kuroki-Suzuki et al., 2007)
BMRI	Human	116		(Trecate et al., 2006)
MR mammography	Human	7	Sensitivity 100%	(Harms et al., 1993)
MR mammography	Human	19	Sensitivity 100%	(Boetes et al., 1997)
MR mammography	Human	36	Sensitivity 94%	(Chan et al., 2010)
MR mammography	Human	8	Sensitivity 87%	(Boetes et al., 1995)
MR mammography	Human	58	Sensitivity 96.5%	(Gilles et al., 1996)
MR mammography	Human	11	Sensitivity 100%	(Soderstrom et al., 1996)
MR mammography	Human	17	Sensitivity 82%	(Bone et al., 1996)
MR mammography	Human	13	Sensitivity 77%	(Orel et al., 1997)
MR mammography	Human	7	Sensitivity 81%	(Kinkel & Vlastos, 2001)
MR mammography	Human	33	Sensitivity 67%	(Westerhof, Fischer, Moritz, & Oestmann, 1998)
MR mammography	Human	50	Sensitivity 58%	(Fischer, Kopka, & Grabbe, 1999)
MR mammography	Human	50	Sensitivity 96%	(Viehweg, Lampe, Buchmann, & Heywang-Köbrunner, 2000)
1^8F-FES-PET	Human	86	Sensitivity 69%	(Dehdashti et al., 1995)
			Specificity 100%	
1^8F-FES-PET	Human	100	Sensitivity 100%	(Mintun et al., 1988)
1^8F-FES-PET	Human	100	Sensitivity 100%	(van Kruchten et al., 2015)
			Specificity 80%	
Spectroscopic Reconstructed Near Infrared Tomographic Imaging	Human	5		(McBride, 2001)
B7-H3–targeted ultrasound	Mice			(Bachawal et al., 2015)
Film- screen mammography	Human	738		(Michell et al., 2012)
Full-field digital mammography	Human	738		(Michell et al., 2012)
Digital breast tomosynthesis	Human	738		(Michell et al., 2012)
The NHS Breast Screening Programme (NHSBSP)	Human	26,475		(Bennett, Sellars, & Moss, 2011)
Digital mammography	Human	49,528	Sensitivity 0.70 ± 0.03	(Pisano et al., 2005)
			Specificity 0.92 ± 0.001	
Film mammography	Human	49,528	Sensitivity 0.66 ± 0.03	(Pisano et al., 2005)
			Specificity 0.92 ± 0.001	
Computer-aided diagnosis	Human	1,083	Sensitivity 81%	(Birdwell, Ikeda, O'Shaughnessy, & Sickles, 2001)
			Specificity 84%	
Digital breast tomosynthesis (DBT)			Sensitivity 69.8%	(Gennaro et al., 2010)
			Specificity 88.9%	
Full-field digital mammography (FFDM)			Sensitivity 74.3%	(Gennaro et al., 2010)
			Specificity 84.8%	
Ultrasonography	Human	241	Sensitivity 100%	(Zanello et al., 2011)

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TABLE 1 (Continued)

Techniques	Model	Samples	Sensitivity and specificity	Ref
			Specificity 89.1%	
Scintimammography	Human	59	Sensitivity 85%	(Das, Biswal, & Bhavaraju, 2006)
			Specificity 90%	
X-ray mammography	Human	59	Sensitivity 89%	(Das et al., 2006)
			Specificity 14%	
Positron emission tomography (PET)	Human	360	Sensitivity 61%	(Wahl, Siegel, Coleman, & Gatsonis, 2004)
			Specificity 80%	
SPECT imaging (MIBI)	Human	146	Sensitivity 96%	(Brem et al., 2008)
			Specificity 59%	
FDG-PET	Human	80	Sensitivity 44%	(Kumar et al., 2006)
			Specificity 95%	
FDG-PET	Human	125	Sensitivity 84.5%	(Gil-Rendo et al., 2006)
			Specificity 98.5%	
Computed tomographic (CT)	Human	29,000		(de González et al., 2009)
Computed tomographic (CT)	Human	1,625		(Chang et al., 1980)

SPECT technique uses isotopes which emit gamma photons, including ^{99}mTc, ^{123}I, or ^{125}I (Benard & Turcotte, 2005).

In a study, Uematsu et al. (2005) assessed the efficacy of PET and SPECT imaging techniques in 15 patients with bone metastases of breast cancer. They results indicated that the sensitivity for diagnosing bone metastases was 85% for SPECT and 17% for PET, specificity of SPECT and PET were 99% and 100%, and accuracy were 96% and 85%, respectively. They revealed that SPECT was classified the bone metastases as 92% of metastases as osteoblastic and 35% of metastases as osteolytic, whereas PET was classified 6% of metastases as osteoblastic and 90% of metastases as osteolytic. These findings suggested that SPECT and PET could be used for diagnosis breast cancer patients with bone metastasis (Uematsu et al., 2005).

Multiple lines of evidence indicated that imaging techniques could provide effective information on breast tumor morphology and functional and metabolic processes within the tumor at various levels (Alcantara et al., 2014). It has been showed that the sensitivity of the various imaging techniques is varied depending on the type of biomarkers applied in signaling the vital processes. These studies suggested that imaging techniques could be employed for diagnosis and monitoring response to therapy in patients with breast cancer (Alcantara et al., 2014).

3 | BIOCHEMICAL MARKERS AND BREAST CANCER

The utilization of various imaging techniques is associated with several limitations such as expensive, lack of sensitivity, specificity (Alcantara et al., 2014). Hence, it seems that introducing new biomarkers which could overcome to the related limitations of imaging techniques are required. The employing of various biomarkers is known as one of important aspects of diagnosis and monitoring patients with breast cancer (Duffy et al., 2017). The suitable biomarkers could contribute to better understanding of cellular and molecular pathways involved in breast cancer pathogenesis (Duffy et al., 2017). These findings could help to designing effective therapeutic approach and monitoring response to treatment in patients with breast cancer. Several studies indicated that a variety of biochemical biomarkers including proteins (i.e., Her2, ER, and Ki67), mRNAs (i.e., ERα, ERβ, and ERRγ), enzymes (i.e., CEA and TSGF), and microRNAs (i.e., miR-21, miR-10b, miR-155, and miR-145) could be used as diagnostic biomarkers for detection and monitoring breast cancer patients (Duffy et al., 2017; Kurozumi et al., 2017) (Table 2).

The human epidermal growth factor receptor 2 (HER2) is one of important proteins involved in breast cancer pathogenesis (Iqbal, 2014). HER2 is found as a member of the epidermal growth factor receptor family which has tyrosine kinase activity (Schrohl, Pedersen, Jensen, Nielsen, & Brunner, 2011). This receptor could be involved in the regulation of cell growth, survival, and differentiation via targeting multiple signal transduction pathways. It has been showed that dimerization of the receptor could lead to auto-phosphorylation of tyrosine residues within the cytoplasmic domain of the receptors (Iqbal, 2014). These events could initiate various cellular and molecular signaling pathways which could contribute to cell proliferation and tumorigenesis (Weigel & Dowsett, 2010). The up regulation and amplification HER2 have been observed in 15–30% cases of breast cancers and 10–30% cases of gastric/gastroesophageal cancers

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TABLE 2 Various biomarkers as predictive biomarkers for breast cancer

Biomarkers	Detection methods	Model	Samples	Ref
Protein				
c-erbB-2 and p53	Immunohistochemistry	Human	90	(Beenken et al., 2001)
BAG-1	Immunohistochemistry	Human	122	(Turner et al., 2001)
ErbB2	Magnetoimmunosensor	Cell line		(Eletxigerra et al., 2015)
	ELISA			
NMP66		Human	100	(Lüftner & Possinger, 2002)
Cox-2	Immunohistochemistry	Human		(Ristimäki et al., 2002)
EZH2	Immunobloting	Human	280	(Kleer et al., 2003)
	Real-Time PCR			
HSP27 (u)	Electroblotting	Human	145	(Rui, Jian-Guo, Yuan-Peng, Hai, & Bing-Gen, 2003)
14-3-3 sigma (d)	Electroblotting	Human	145	(Rui et al., 2003)
Her-2/neu	ELISA	Human	488	(Ferrero-Poüs et al., 2000)
Her-2/neu	ELISA	Human	1,123	(Eppenberger-Castori et al., 2001)
ERα	Western blot	Human	242	(Fuqua et al., 2003)
ERβ	Western blot	Human	242	(Fuqua et al., 2003)
HER2	Immunohistochemistry	Human	506	(Yamashita et al., 2003)
CD24	Immunohistochemistry	Human	201	(Kristiansen et al., 2003)
HNP1	ELISA	Human	95	(Li et al., 2005)
HNP2	ELISA	Human	95	(Li et al., 2005)
HNP3	ELISA	Human	95	(Li et al., 2005)
Protein kinase Cε (PKCε)	Immunohistochemistry	Cell line		(Pan et al., 2005)
	Western blot	In vivo		
GATA3	Western blot	Human	139	(Mehra et al., 2005)
HCCR	ELISA	Human	299	(Jung et al., 2005)
caPCNA isoform	Immunohistochemistry	Cell line		(Malkas et al., 2006)
	Western blot			
Metallothionein	Immunohistochemistry	Cell line		(Bay, Jin, Huang, & Tan, 2006)
	Western blot			
Leptin and ObR	Immunohistochemistry	Human	304	(Garofalo et al., 2006)
BC1	Protein Chip array	Human	89	(Mathelin, Cromer, Wendling, Tomasetto, & Rio, 2006)
BC3	Protein Chip array	Human	89	(Mathelin et al., 2006)
Ki-67		Human		(De Azambuja et al., 2007)
Matrix metalloproteinases (MMP)	Zymography	Human	148	(Pories et al., 2008)
	Immunobloting			
ADAM 12	Western blots	Human	148	(Pories et al., 2008)
PDGF	Radioimmunoassay	Human	110	(Pasanisi et al., 2008)
IGF-I	Radioimmunoassay	Human	110	(Pasanisi et al., 2008)
CA 15-3	Electrochemiluminescent immunoenzymometric assay	Human	41	(Toth et al., 2008)
LG3 fragment	Western blot analysis selected reaction monitoring (SRM)	Human	6	(Chang et al., 2008)
		Cell line		
TBX3	Immunohistochemistry	Human	42	(Yarosh et al., 2008)

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TABLE 2 (Continued)

Biomarkers	Detection methods	Model	Samples	Ref
		Cell line		
SPAG9	ELISA	Human	100	(Kanojia, Garg, Gupta, Gupta, & Suri, 2009)
	Immunohistochemistry	Cell line		
AHSG anti-AHSG autoantibody	Western blot	Human	81	(Yi et al., 2009)
Activated leukocyte cell adhesion molecule (ALCAM)	quantitative ELISA	Human	150	(Doolan et al., 2009)
EFEMP1	Immunohistochemistry	Human		(Sadr-Nabavi et al., 2009)
Ki-67	Immunohistochemistry	Human	107	(Soliman & Yussif, 2016b)
CA15-3	Electrochemiluminescence	Human	86	(Wang et al., 2014)
CA125	Electrochemiluminescence	Human	86	(Wang et al., 2014)
CEA	Electrochemiluminescence	Human	86	(Wang et al., 2014)
TSGF	Electrochemiluminescence	Human	86	(Wang et al., 2014)
mRNA				
Her-2/neu	Real-Time PCR	Human	365	(Pawlowski, Révillion, Hebbar, Hornez, & Peyrat, 2000)
ERRα	Real-Time PCR	Human	38	(Ariazi, Clark, & Mertz, 2002)
ERRγ	Real-Time PCR	Human	38	(Ariazi et al., 2002)
SPAG9	Real-Time PCR	Human	100	(Kanojia et al., 2009)
		Cell line		
TMEM25	qRT-PCR	Human		(Doolan et al., 2009)
REPS2	qRT-PCR	Human		(Doolan et al., 2009)
Meis 1	qRT-PCR	Human		(Doolan et al., 2009)
MCM5	qRT-PCR	Human		(Eissa, Matboli, Shehata, & Essawy, 2015)
Gene				
C35 (C17orf37)	Northern blot and real-time PCR	Human		(Evans et al., 2006)
Phosphocholine	Phosphocholine cytidylyltransferase (CCT) assay	Cell line		(Eliyahu, Kreizman, & Degani, 2007)
LINE1	Real-Time PCR	Human	36	(Sunami, Vu, Nguyen, Giuliano, & Hoon, 2008)
ESR1	Methylation specific PCR	Human	107	(Martínez-Galán et al., 2008)
14-3-3-σ gene	Methylation specific PCR	Human	107	(Martínez-Galán et al., 2008)
Free Circulating Tumor DNA	Real-Time PCR	Human	175	(Catarino et al., 2008)
ccf nDNA	Multiplex real-time PCR	Human	148	(Kohler et al., 2009)
mtDNA	Multiplex real-time PCR	Human	148	(Kohler et al., 2009)
EFEMP1	Real-Time PCR	Human		(Sadr-Nabavi et al., 2009)
	Microarray			
KLK10 exon 3 methylation	Methylation-specific PCR	Human	35	(Kioulafa et al., 2009)
HOTAIR	Real-Time PCR	Human	24	(Zhang et al., 2015)
KLF4	Tissue microarray (TMA)	Human	208	(Nagata et al., 2017)
NANOG	Tissue microarray (TMA)	Human	208	(Nagata et al., 2017)
MicroRNA				
miR-125b	Microarray	Human	76	(Iorio et al., 2005)

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TABLE 2 (Continued)

Biomarkers	Detection methods	Model	Samples	Ref
	Northern blot	Cell line		
miR-145	Microarray	Human	76	(Iorio et al., 2005)
	Northern blot	Cell line		
miR-21	Microarray	Human	76	(Iorio et al., 2005)
	Northern blot	Cell line		
miR-155	Microarray	Human	76	(Iorio et al., 2005)
	Northern blot	Cell line		
miR-29b-2	Microarrays	Cell lines		(Mattie et al., 2006)
	qRT-PCR	Human		
miR-17-5p	Microarrays	Cell lines		(Mattie et al., 2006)
	qRT-PCR	Human		
miR-27b	Microarrays	Cell lines		(Mattie et al., 2006)
	qRT-PCR	Human		
miR-10b	Microarrays	Cell lines		(Mattie et al., 2006)
	qRT-PCR	Human		
miR-195	qRT-PCR	Human	83	(Heneghan et al., 2010)
let-7a	qRT-PCR	Human	83	(Heneghan et al., 2010)
miR-595 (u)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-589 (u)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-425 (u)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-155 (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-340 (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-181a (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-181b (d)	qRT-PCR	Human	63	(Sochor et al., 2014)
miR-1275 (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-1304 (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-151-5p (d)	qRT-PCR	Human	40	(Zhao et al., 2010)
miR-197	qRT-PCR	Human	180	(Shaker, Maher, Nassar, Morcos, & Gad, 2015)
miR-205	qRT-PCR	Human	180	(Shaker et al., 2015)
miRNA-10b	qRT-PCR	Human		(Eissa et al., 2015)
miR-183/182/96 cluster	qRT-PCR	Human		(Song et al., 2016)
miR-148b	qRT-PCR	Human	50	(Shen et al., 2014)
		Cell line		
miR-133a	qRT-PCR	Human	50	(Shen et al., 2014)
		Cell line		
miR-19a	qRT-PCR	Human	63	(Sochor et al., 2014)
miR-24 (d)	qRT-PCR	Human	63	(Sochor et al., 2014)
miR-1280	Microarray	Human	11	(Park et al., 2014)
miR-1260	Microarray	Human	11	(Park et al., 2014)
miR-720	Microarray	Human	11	(Park et al., 2014)

(Iqbal, 2014). Hence, HER2 could be used as predictive biomarkers for patients with breast cancer.

Ki-67 is other important protein which plays critical roles in the breast cancer pathogenesis. This protein is known as a nuclear protein which is associated with cellular proliferation (Scholzen & Gerdes, 2000). It has been showed that Ki-67 nuclear antigen could be expressed in certain phases of the cell cycle including S, G1, G2, and M phases, but is non-existing in G0 (Gerdes et al., 1984; Scholzen &

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Gerdes, 2000). Several studies indicated that Ki-67 could be involved in breast cancer pathogenesis.

In a study, Soliman and Yussif (2016a) assessed the utilization of Ki-67 as prognostic and diagnostic biomarkers for monitoring 107 patients with breast cancer. They also investigated the estrogen receptor, progesterone receptor, and HER2 in patients with breast cancer. Their results indicated that 44, 23, 15, and 25 subjects were classified as luminal A, luminal B, HER2 subtype, and triple-negative (TN), respectively. They showed that in No luminal A patients levels of Ki-67 were higher than 15%, and their recurrence was 20%. In luminal B group, Ki-67 levels were higher than 15% and recurrence was 39% in 69% of patients. In HER2 subtype, in 34% of cases, Ki-67 levels were higher than 15% and recurrence was 40%. Finally, in triple-negative subjects, in 60% of cases, Ki-67 levels were higher than 15%, and recurrence was observed in 32% of patients. These findings suggested that levels of Ki-67 could be applied as predictive biomarkers for monitoring patients with breast cancer (Soliman & Yussif, 2016a).

Besides HER2, estrogen receptor (ER), and progesterone receptor (PR) could be used for classification of breast tumor subtyping (Dai, Xiang, Li, & Bai, 2016). The steroid hormone, estradiol, have critical roles in the initiation and progression of breast cancer. It has been showed that various breast cancers could start out as estrogen dependent and express the ER (Dai et al., 2016). Among of various subtypes of ER, ERα is known as main ER subtypes in the mammary epithelium which has important tasks in mammary gland biology and in progression of breast cancer (Li et al., 2010). The following binding of estrogen to ERα, the ligand-activated ERα could be translocated to the nucleus and bind to the responsive element in the promoter. These actions via promoting gene transcription could be associated with stating of a variety of cellular and molecular pathways involved in the progression of breast cancer (McDonnell & Norris, 2002). Multiple lines of evidence indicated that estrogen and ER could contribute to metastasis of breast cancer. Estrogen signaling could affect on epithelial-to-mesenchymal transition (EMT) and ERα signaling has interaction with various EMT regulators such as Snail and Slug (Saha Roy & Vadlamudi, 2012). ERα could bind to the promoter of metastasis tumor antigen (MTA) 3 which could inhibit Snail. Snail is known as a gene which could affect on EMT transition (Fujita et al., 2003). ERα are able to regulation of Slug transcription via making of a co-repressor complex including HDAC1 (histone deacetylase 1) and N-CoR (nuclear receptor co-repressor) (Ye et al., 2008). ERβ, similar to ERα could be involved in estrogen signaling. Evolving evidence indicated that up regulation of ERβ could decrease cell proliferation and it could be had anti-proliferative (tumor suppressor) functions (Speirs et al., 2004; Strom et al., 2004). Down regulation of ERβ was observed in metastatic breast cancer, and up regulation of ERβ expression could be related with less metastatic condition and proliferating tumors (Jarvinen, Pelto-Huikko, Holli, & Isola, 2000; Skliris et al., 2003). These findings indicated that various subtypes of ER (i.e., ERα and ERβ) could be employed as effective predictive biomarkers for detection and monitoring patients with breast cancer. Table 2 listed various predictive biomarkers for detection and monitoring patients with breast cancer.

4 | MIcroRNAs AND EXOSOMES AS DIAGNOSTIC BIOMARKERS FOR BREAST CANCER

MicroRNAs (miRNAs) are known as small non-coding RNAs that involved in the regulation of the various target genes at the post-transcriptional phase (Fathullahzadeh, Mirzaei, Honardoost, Sahebkar, & Salehi, 2016; Golabchi et al., 2017; Masoudi, Mehrabian, & Mirzaei, 2017; Mirzaei, Masoudifar, et al., 2017; Simonian, Mosallayi, & Mirzaei, 2017). These molecules act as epigenetic regulators and play critical roles in various biological processes such as development, angiogenesis, growth and differentiation (Hashemi Goradel et al., 2017; Mashreghi et al., 2017; Mirzaei et al., 2016; Rabieian et al., 2017; Rashidi, Malekzadeh, Goodarzi, Masoudifar, & Mirzaei, 2017). It has been showed that deregulation of miRNAs are related with initiation and progression of various diseases such as cancer, cardiovascular, stroke, and diabetes (Hoseini et al., 2017; Mirzaei, 2017; Mirzaei, Ferns, Avan, & Mobarhan, 2017; Mirzaei, Momeni, et al., 2017; Mohammadi, Goodarzi, Jaafari, Mirzaei, & Mirzaei, 2016). Evolving evidence indicated that miRNAs via targeting a variety of cellular and molecular pathways could be involved in breast cancer pathogenesis (Kurozumi et al., 2017). Recent reports showed that expression profiles of miRNAs could be associated with tumor aggressiveness, response to treatment and patient outcome in breast cancer (Zhang et al., 2017). One of important aspects of breast cancer is the identification of intrinsic subtype of the primary tumor. It has been showed that for each intrinsic subtype of breast tumors must be used specific treatment regimen. Several reports revealed that each intrinsic subtypes of breast cancer have different miRNA expression profiles (Kurozumi et al., 2017) (Table 2). Hence, it seems that the assessment of expression profiles of miRNAs could be used as effective candidates for diagnosis and monitoring patients with breast cancer.

In a study, Park et al., investigated a panel of circulating miRNAs as diagnostic biomarkers for monitoring 11 patients with ER positive breast cancer (ER+/HER2−) using microarray and qRT-PCR (Park et al., 2014). Moreover, they assessed the relationship between the expression levels of circulating miRNAs and clinical outcomes of ER positive metastatic breast cancer. Their results indicated that expression levels of various miRNAs including miR-1280, miR-1260, and miR-720 were up-regulated in blood samples of cancer patients. Moreover, they showed that Among 37 metastatic breast cancer patients, miR-1280 (known as a circulating tRNA-derived miRNA) levels significantly were down regulated after treatment in patients who received systemic therapy. These results proposed that expression profiles of various miRNAs could be used as diagnostic biomarkers for patients with breast cancer (Park et al., 2014). It has been showed that a variety of miRNAs including Let-7, miR-34c, miR-200c, miR-183, miR-16, and miR-203 could be involved in various stages of breast cancer such as breast tumor-initiating cells via affecting on several

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cellular and molecular targets (i.e., Sp1, Wip1, EMT, and BMI1) (Shimono et al., 2009; Yu et al., 2012; Zhang et al., 2010).

In a study, Shimono et al. (2009) assessed the expression levels of various miRNAs in breast cancer stem cell populations and normal stem cells. They results indicated that 37 miRNAs have different expression profiles in breast cancer stem cells than normal stem cells. They showed that expression levels of three clusters including miR-183-96-182, miR-200c-141, and miR-200b-200a-429 were significantly decreased in human breast cancer stem cells than normal stem cells. Moreover, they showed that miR-200c could regulate expression of BMI1 which known as a regulator of stem cell self-renewal. MiR-200c is able to inhibit the clonal expansion of breast cancer cells and inhibit the growth of embryonal carcinoma cells in vitro. It has been showed that miR-200c could prevent of tumor formation driven by human breast cancer stem cells in vivo. These findings that deregulation of these miRNAs involved in initiation and progression of breast cancer and might be used as diagnostic and therapeutic biomarkers for breast cancer patients (Shimono et al., 2009).

The miR-200 family is known as one of important miRNAs which involved in various biological processes such as self-renewal, and metastasis. It has been showed that this cluster via affecting on EMT could contribute to initiation of metastasis in breast cancer patients. Previously studies revealed that miR-200c via targeting two pathways including p53, and ZEB1-miR-200c-BMI1 could anticipate in metastasis of breast cancer cells (Ahmad et al., 2011; Chang et al., 2011).

In a study, Zhang et al. (2017) assessed the expression profiles of various miRNAs in metastatic breast cancer. Their results showed that the plasma levels of miR-200c and miR-141 were up regulated in Foxp3 heterozygous Scurfy mutant. They confirmed that expression of miR-200c and miR-141 could be modulated by a FOXP3-KAT2B axis in breast cancer cells, and plasma levels of miR-200c and miR-141 could be used as diagnostic biomarkers for detection patients with early stage of breast cancer metastasis (Zhang et al., 2017).

Besides miR-200 family, deregulation of other miRNAs such as miR-34c, miR-106b-25 cluster, miR-30a, and miR-30c via targeting various cellular and molecular involved in EMT (e.g., TWF1, VIM) could contribute to metastasis in breast cancer cells (Bockhorn et al., 2013; Cheng et al., 2012; Smith et al., 2012; Yu et al., 2012).

These studies confirmed that deregulation of various miRNAs play key roles in initiation and progression of breast cancer. Hence, it seems that utilization of these miRNAs could be introduced as suitable diagnostic and therapeutic biomarkers for patients with breast cancer.

Exosomes are other candidates which could be employed as diagnostic and therapeutic biomarkers for detection and monitoring of patients with breast cancer (Jia et al., 2017). Exosomes are known as biological nano-vehicles which could transfer a variety of cellular and molecular signals to recipient cells (Banikazemi et al., 2017; Saeedi Borujeni et al., 2017). These signals have major roles in various physiological events (Mirzaei, Sahebkar, Jaafari, Goodarzi, & Mirzaei, 2017; Moradian Tehrani et al., 2017). Exosomes transfer their signals via targeting their cargos (i.e., proteins, mRNAs, and miRNAs) to recipient cells (Mirzaei, Sahebkar, et al., 2017; Saadatpour, Fadaee, et al., 2016). Multiple lines of evidence revealed that exosomes and their cargos could be involved in pathogenesis of various cancers such as breast cancer (Jia et al., 2017). Hence, it seems that exosomes and their cargos could be used as diagnostic and therapeutic biomarkers in breast cancer.

Among of various cargos which could be carried by exosomes, miRNAs have been emerged as major cargos with big roles in patients with breast cancer (Sempere, Keto, & Fabbri, 2017). It has been showed that exosomal miRNAs have critical function in various diseases such as breast cancer (Sempere et al., 2017).

In a study, Hannafon et al. (2016) assessed the plasma levels of various exosomal miRNAs in patients with breast cancer. Their results showed that exosomes released from breast cancer cells are enriched with various miRNAs including miR-21 and miR-1246. The evaluation of plasma levels of these exosomal miRNAs showed that these molecules were up regulated in plasma of patients with breast cancer. These findings suggested that exosomal miRNAs could be used as diagnostic candidates for detection and monitoring patients with breast cancer (Hannafon et al., 2016).

In other study, Singh, Pochampally, Watabe, Lu, and Mo (2014) indicated that exosomal miR-10b could induce metastasis in breast cancer cells. Their results revealed that exosomal miRNAs could be carried among different cell lines by direct uptake. They showed that exosomal miR-10b was up regulated in MDA-MB-231 cells (as metastatic breast cancer) than non-metastatic breast cancer cells or normal breast cells. Moreover, they indicated that nSMase2 or ceramide could induce the secretion of exosomal miR-10b whereas ceramide inhibitor could inhibit this action. miR-10b are able to inhibit the expression of its target genes (e.g., HOXD10 and KLF4) which play key roles in breast cancer metastasis. These results proposed that exosomal miRNAs relaeased from metastatic breast cancer cells could have critical roles in progression of metastasis in breast cancer cells. Hence, exosomal miRNAs could be introduced as new diagnosis and therapeutic options for patients with breast cancer (Singh et al., 2014).

Developmental Endothelial Locus-1 (Del-1) is one of important proteins involved in cancer (Beckham et al., 2014). This proteins is known as an extracellular matrix protein which has three N-terminal epidermal growth factor-like domains and factor V C domains, C1, and C2 (Schurpf et al., 2012). It has been showed that this protein could express by endothelial cells during embryonic vascular development and induces the adhesion of endothelial cells via its interaction with other proteins such as integrin receptors (Hidai et al., 1998; Schurpf et al., 2012). Several studies indicated that Del-1 could be expressed in various cancers such as breast cancer (Aoka et al., 2002; Sun et al., 2010). These results proposed that Del-1 could be used as cancer-specific biomarker for various human cancers such as breast cancer. Recently, in a study, Chae et al. (2015) assessed the expression levels of exosomal Del-1 in 10 patients with early stages of breast cancer (stages I and II). Their results revealed that the levels of exosomal Del-1 was significantly up regulated in patients with breast cancer at all stages than other groups. These findings suggested that exosomal Del-1 could be utilized as new diagnostic biomarker for patients with breast cancer (Chae et al., 2015).

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5 | CONCLUSION

Breast cancer is one of important cancers among women which is known as the second most common cancer worldwide. Diagnosis of breast cancer is main aspects of breast cancer therapy. Multiple lines of evidence indicated that a variety of approaches and biomarkers have potential to are used as diagnostic approaches for detection and monitoring patients with breast cancer. Imaging techniques including mammography, MRI, SPECT, PET, and CT and development of them could be employed for diagnosis and monitoring breast cancer patients. Despite, more development of imaging techniques, utilization of them are associated with various limitations such as expensive and sensitivity. Hence, it seems that identification new tools for diagnosis of breast cancer patients are required. It has been showed that utilization of new biomarkers such as evaluation of expression levels of various proteins (i.e., ER, Ki67, PR, and HER2) and various molecules (miRNAs and exosomes) have opened new opportunity for diagnosis and monitoring patients with breast cancer.

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REFERENCES

Ahmad, A., Aboukameel, A., Kong, D., Wang, Z., Sethi, S., Chen, W., et al. (2011). Phosphoglucose isomerase/autocrine motility factor mediates epithelial-mesenchymal transition regulated by miR-200 in breast cancer cells. *Cancer Research*, *71*(9), 3400–3409.

Alcantara, D., Leal, M. P., Garcia-Bocanegra, I., & Garcia-Martin, M. L. (2014). Molecular imaging of breast cancer: Present and future directions. *Frontiers in Chemistry*, *18*(2), 112.

Andreea, G. I., Pegza, R., Lascu, L., Bondari, S., Stoica, Z., & Bondari, A. (2011). The role of imaging techniques in diagnosis of breast cancer. *Journal of Current Health Sciences*, *37*(2), 241–248.

Aoka, Y., Johnson, F. L., Penta, K., Hirata Ki, K., Hidai, C., Schatzman, R., et al. (2002). The embryonic angiogenic factor Del1 accelerates tumor growth by enhancing vascular formation. *Microvascular Research*, *64*(1), 148–161.

Ariazi, E. A., Clark, G. M., & Mertz, J. E. (2002). Estrogen-related receptor α and estrogen-related receptor γ associate with unfavorable and favorable biomarkers, respectively, in human breast cancer. *Cancer Research*, *62*(22), 6510–6518.

Bachawal, S. V., Jensen, K. C., Wilson, K. E., Tian, L., Lutz, A. M., & Willmann, J. K. (2015). Breast cancer detection by B7-H3-targeted ultrasound molecular imaging. *Cancer Research*, *75*(12), 2501–2509.

Banikazemi, Z., Haji, H. A., Mohammadi, M., Taheripak, G., Iranifar, E., Poursadeghiyan, M., et al. (2017). Diet and cancer prevention: Dietary compounds, dietary MicroRNAs and dietary exosomes. *Journal of Cellular Biochemistry*, *28*(10), 26244.

Barber, M. D., Jack, W., & Dixon, J. M. (2004). Diagnostic delay in breast cancer. *British Journal of Surgery*, *91*(1), 49–53.

Bay, B.-H., Jin, R., Huang, J., & Tan, P.-H. (2006). Metallothionein as a prognostic biomarker in breast cancer. *Experimental Biology and Medicine*, *231*(9), 1516–1521.

Beckham, C. J., Olsen, J., Yin, P. N., Wu, C. H., Ting, H. J., Hagen, F. K., et al. (2014). Bladder cancer exosomes contain EDIL-3/Del1 and facilitate cancer progression. *Journal of Urology*, *192*(2), 583–592.

Beenken, S. W., Grizzle, W. E., Crowe, D. R., Conner, M. G., Weiss, H. L., Sellers, M. T., et al. (2001). Molecular biomarkers for breast cancer prognosis: Coexpression of c-erbB-2 and p53. *Annals of Surgery*, *233*(5), 630.

Benard, F., & Turcotte, E. (2005). Imaging in breast cancer: Single-photon computed tomography and positron-emission tomography. *Breast Cancer Research*, *7*(4), 153–162.

Bennett, R., Sellars, S., & Moss, S. (2011). Interval cancers in the NHS breast cancer screening programme in England, Wales and Northern Ireland. *British Journal of Cancer*, *104*(4), 571.

Birdwell, R. L., Ikeda, D. M., O'Shaughnessy, K. F., & Sickles, E. A. (2001). Mammographic characteristics of 115 missed cancers later detected with screening mammography and the potential utility of computer-aided detection. *Radiology*, *219*(1), 192–202.

Bockhorn, J., Yee, K., Chang, Y. F., Prat, A., Huo, D., Nwachukwu, C., et al. (2013). MicroRNA-30c targets cytoskeleton genes involved in breast cancer cell invasion. *Breast Cancer Research and Treatment*, *137*(2), 373–382.

Boetes, C., Mus, R., Holland, R., Barentsz, J. O., Strijk, S. P., Wobbes, T., et al. (1995). Breast tumors: Comparative accuracy of MR imaging relative to mammography and US for demonstrating extent. *Radiology*, *197*(3), 743–747.

Boetes, C., Strijk, S., Holland, R., Barentsz, J., Van Der Sluis, R., & Ruijs, J. (1997). False-negative MR imaging of malignant breast tumors. *European Radiology*, *7*(8), 1231–1234.

Bone, B., Aspelin, P., Bronge, L., Isberg, B., Perbeck, L., & Veress, B. (1996). Sensitivity and specificity of MR mammography with histopathological correlation in 250 breasts. *Acta Radiologica*, *37*(2), 208–213.

Brem, R. F., Floerke, A. C., Rapelyea, J. A., Teal, C., Kelly, T., & Mathur, V. (2008). Breast-specific gamma imaging as an adjunct imaging modality for the diagnosis of breast cancer. *Radiology*, *247*(3), 651–657.

Catarino, R., Ferreira, M. M., Rodrigues, H., Coelho, A., Nogal, A., Sousa, A., et al. (2008). Quantification of free circulating tumor DNA as a diagnostic marker for breast cancer. *DNA and Cell Biology*, *27*(8), 415–421.

Chae, Y. S., Moon, P.-G., Lee, J.-E., Cho, Y.-E., Lee, S. J., Jung, J. H., et al. (2015). Exosome-derived developmental endothelial locus-1 as a novel diagnostic marker for breast cancer. *Journal of Clinical Oncology*, *33*(S15), https://doi.org/10.1200/jco.2015.33.15_suppl.e22262.

Chan, S., Chen, J.-H., Agrawal, G., Lin, M., Mehta, R. S., Carpenter, P. M., et al. (2010). Characterization of pure ductal carcinoma in situ on dynamic contrast-enhanced MR imaging: Do nonhigh grade and high grade show different imaging features? *Journal of Oncology*, *2010*, pii: 431341.

Chang, C., Sibala, J. L., Fritz, S. L., Dwyer, S. J., Templeton, A. W., Lin, F., et al. (1980). Computed tomography in detection and diagnosis of breast cancer. *Cancer*, *46*(S4), 939–946.

Chang, J. W., Kang, U. B., Kim, D. H., Yi, J. K., Lee, J. W., Noh, D. Y., et al. (2008). Identification of circulating endorepellin LG3 fragment: Potential use as a serological biomarker for breast cancer. *PROTEOMICS-Clinical Applications*, *2*(1), 23–32.

Chang, C. J., Chao, C. H., Xia, W., Yang, J. Y., Xiong, Y., Li, C. W., et al. (2011). P53 regulates epithelial-mesenchymal transition and stem cell properties through modulating miRNAs. *Nature Cell Biology*, *13*(3), 317–323.

Cheng, C. W., Wang, H. W., Chang, C. W., Chu, H. W., Chen, C. Y., Yu, J. C., et al. (2012). MicroRNA-30a inhibits cell migration and invasion by downregulating vimentin expression and is a potential prognostic marker in breast cancer. *Breast Cancer Research and Treatment*, *134*(3), 1081–1093.

Dai, X., Xiang, L., Li, T., & Bai, Z. (2016). Cancer hallmarks, biomarkers and Breast cancer molecular subtypes. *Journal of Cancer*, *7*(10), 1281–1294.

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Das, B. K., Biswal, B. M., & Bhavaraju, M. (2006). Role of scintimammography in the diagnosis of breast cancer. *The Malaysian Journal of Medical Sciences: MJMS*, *13*(1), 52.

De Azambuja, E., Cardoso, F., de Castro Jr, G., Colozza, M., Mano, M. S., Durbecq, V., et al. (2007). Ki-67 as prognostic marker in early breast cancer: A meta-analysis of published studies involving 12 155 patients. *British Journal of Cancer*, *96*(10), 1504.

de González, A. B., Mahesh, M., Kim, K.-P., Bhargavan, M., Lewis, R., Mettler, F., et al. (2009). Projected cancer risks from computed tomographic scans performed in the United States in 2007. *Archives of Internal Medicine*, *169*(22), 2071–2077.

Dehdashti, F., Mortimer, J. E., Siegel, B. A., Griffeth, L. K., Bonasera, T. J., Fusselman, M. J., et al. (1995). Positron tomographic assessment of estrogen receptors in breast cancer: Comparison with FDG-PET and in vitro receptor assays. *Journal of Nuclear Medicine*, *36*(10), 1766–1774.

Dheeba, J., Singh, N. A., & Selvi, S. T. (2014). Computer-aided detection of breast cancer on mammograms: A swarm intelligence optimized wavelet neural network approach. *Journal of Biomedical Informatics*, *49*, 45–52.

Doolan, P., Clynes, M., Kennedy, S., Mehta, J., Germano, S., Ehrhardt, C., et al. (2009). TMEM25, REPS2 and Meis 1: Favourable prognostic and predictive biomarkers for breast cancer. *Tumor Biology*, *30*(4), 200–209.

Duffy, M. J., Harbeck, N., Nap, M., Molina, R., Nicolini, A., Senkus, E., et al. (2017). Clinical use of biomarkers in breast cancer: Updated guidelines from the European Group on Tumor Markers (EGTM). *European Journal of Cancer*, *75*, 284–298.

Eissa, S., Matboli, M., Shehata, H. H., & Essawy, N. O. (2015). MicroRNA-10b and minichromosome maintenance complex component 5 gene as prognostic biomarkers in breast cancer. *Tumor Biology*, *36*(6), 4487–4494.

Eletxigerra, U., Martinez-Perdiguero, J., Merino, S., Barderas, R., Torrente-Rodríguez, R., Villalonga, R., et al. (2015). Amperometric magneto-immunosensor for ErbB2 breast cancer biomarker determination in human serum, cell lysates and intact breast cancer cells. *Biosensors and Bioelectronics*, *70*, 34–41.

Eliyahu, G., Kreizman, T., & Degani, H. (2007). Phosphocholine as a biomarker of breast cancer: Molecular and biochemical studies. *International Journal of Cancer*, *120*(8), 1721–1730.

Eppenberger-Castori, S., Kueng, W., Benz, C., Caduff, R., Varga, Z., Bannwart, F., et al. (2001). Prognostic and predictive significance of ErbB-2 breast tumor levels measured by enzyme immunoassay. *Journal of Clinical Oncology*, *19*(3), 645–656.

Evans, E. E., Henn, A. D., Jonason, A., Paris, M. J., Schiffhauer, L. M., Borrello, M. A., et al. (2006). C35 (C17orf37) is a novel tumor biomarker abundantly expressed in breast cancer. *Molecular Cancer Therapeutics*, *5*(11), 2919–2930.

Fathullahzadeh, S., Mirzaei, H., Honardoost, M. A., Sahebkar, A., & Salehi, M. (2016). Circulating microRNA-192 as a diagnostic biomarker in human chronic lymphocytic leukemia. *Cancer Gene Therapy*, *23*(10), 327–332.

Ferrero-Poüs, M., Hacène, K., Bouchet, C., Le Doussal, V., Tubiana-Hulin, M., & Spyratos, F. (2000). Relationship between c-erbB-2 and other tumor characteristics in breast cancer prognosis. *Clinical Cancer Research*, *6*(12), 4745–4754.

Fischer, U., Kopka, L., & Grabbe, E. (1999). Breast carcinoma: Effect of preoperative contrast-enhanced MR imaging on the therapeutic approach. *Radiology*, *213*(3), 881–888.

Fujita, N., Jaye, D. L., Kajita, M., Geigerman, C., Moreno, C. S., & Wade, P. A. (2003). MTA3, a Mi-2/NuRD complex subunit, regulates an invasive growth pathway in breast cancer. *Cell*, *113*(2), 207–219.

Fuqua, S. A., Schiff, R., Parra, I., Moore, J. T., Mohsin, S. K., Osborne, C. K., et al. (2003). Estrogen receptor β protein in human Breast cancer. *Cancer Research*, *63*(10), 2434–2439.

Garofalo, C., Koda, M., Cascio, S., Sulkowska, M., Kanczuga-Koda, L., Golaszewska, J., et al. (2006). Increased expression of leptin and the leptin receptor as a marker of breast cancer progression: Possible role of obesity-related stimuli. *Clinical Cancer Research*, *12*(5), 1447–1453.

Gennaro, G., Toledano, A., Di Maggio, C., Baldan, E., Bezzon, E., La Grassa, M., et al. (2010). Digital breast tomosynthesis versus digital mammography: A clinical performance study. *European Radiology*, *20*(7), 1545–1553.

Gerdes, J., Lemke, H., Baisch, H., Wacker, H. H., Schwab, U., & Stein, H. (1984). Cell cycle analysis of a cell proliferation-associated human nuclear antigen defined by the monoclonal antibody Ki-67. *Journal of Immunology*, *133*(4), 1710–1715.

Gholamin, S., Mirzaei, H., Razavi, S. M., Hassanian, S. M., Saadatpour, L., Masoudifar, A., et al. (2017). GD2-targeted immunotherapy and potential value of circulating microRNAs in neuroblastoma. *Journal of Cellular Physiology*, *1*(10), 25793.

Gil-Rendo, A., Zornoza, G., García-Velloso, M., Regueira, F., Beorlegui, C., & Cervera, M. (2006). Fluorodeoxyglucose positron emission tomography with sentinel lymph node biopsy for evaluation of axillary involvement in breast cancer. *British Journal of Surgery*, *93*(6), 707–712.

Gilles, R., Meunier, M., Lucidarme, O., Zafrani, B., Guinebretière, J.-M., Tardivon, A. A., et al. (1996). Clustered breast microcalcifications: Evaluation by dynamic contrast-enhanced subtraction MRI. *Journal of Computer Assisted Tomography*, *20*(1), 9–14.

Golabchi, K., Soleimani-Jelodar, R., Aghadoost, N., Momeni, F., Moridikia, A., Nahand, J. S., et al. (2017). MicroRNAs in retinoblastoma: Potential diagnostic and therapeutic biomarkers. *Journal of Cellular Physiology*, *28*(10), 26070.

Hannafon, B. N., Trigoso, Y. D., Calloway, C. L., Zhao, Y. D., Lum, D. H., Welm, A. L., et al. (2016). Plasma exosome microRNAs are indicative of breast cancer. *Breast Cancer Research*, *18*(1), 016–0753.

Harms, S., Flamig, D., Hesley, K., Meiches, M., Jensen, R., Evans, W., et al. (1993). MR imaging of the breast with rotating delivery of excitation off resonance: Clinical experience with pathologic correlation. *Radiology*, *187*(2), 493–501.

Hashemi Goradel, N., Ghiyami-Hour, F., Jahangiri, S., Negahdari, B., Sahebkar, A., Masoudifar, A., et al. (2017). Nanoparticles as new tools for inhibition of cancer angiogenesis. *Journal of Cellular Physiology*, *25*(10), 26029.

Heneghan, H. M., Miller, N., Lowery, A. J., Sweeney, K. J., Newell, J., & Kerin, M. J. (2010). Circulating microRNAs as novel minimally invasive biomarkers for breast cancer. *Annals of Surgery*, *251*(3), 499–505.

Hidai, C., Zupancic, T., Penta, K., Mikhail, A., Kawana, M., Quertermous, E. E., et al. (1998). Cloning and characterization of developmental endothelial locus-1: An embryonic endothelial cell protein that binds the alphavbeta3 integrin receptor. *Genes & Development*, *12*(1), 21–33.

Hoseini, Z., Sepahvand, F., Rashidi, B., Sahebkar, A., Masoudifar, A., & Mirzaei, H. (2017). NLRP3 inflammasome: Its regulation and involvement in atherosclerosis. *Journal of Cellular Physiology*, *27*(10), 25930.

Iorio, M. V., Ferracin, M., Liu, C.-G., Veronese, A., Spizzo, R., Sabbioni, S., et al. (2005). MicroRNA gene expression deregulation in human breast cancer. *Cancer Research*, *65*(16), 7065–7070.

Iqbal, N. (2014). Human epidermal growth factor receptor 2 (HER2) in cancers: Overexpression and therapeutic implications. *Molecular Biology International*, *852748*(10), 7.

Jarvinen, T. A., Pelto-Huikko, M., Holli, K., & Isola, J. (2000). Estrogen receptor beta is coexpressed with ERalpha and PR and associated with nodal status, grade, and proliferation rate in breast cancer. *American Journal of Pathology*, *156*(1), 29–35.

Jia, W. R., Chai, W. M., Tang, L., Wang, Y., Fei, X. C., Han, B. S., et al. (2014). Three-dimensional contrast enhanced ultrasound score and dynamic contrast-enhanced magnetic resonance imaging score in evaluating breast tumor angiogenesis: Correlation with biological factors. *European Journal of Radiology*, *83*(7), 1098–1105.

Jia, Y., Chen, Y., Wang, Q., Jayasinghe, U., Luo, X., Wei, Q., et al. (2017). Exosome: Emerging biomarker in breast cancer. *Oncotarget*, *8*(25), 41717–41733.

10974652, 2018, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jcp.26379 by Central University Of Rajasthan, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Jung, S. S., Park, H. S., Lee, I. J., Namkoong, H., Shin, S. M., Cho, G. W., et al. (2005). The HCCR oncoprotein as a biomarker for human breast cancer. *Clinical Cancer Research*, *11*(21), 7700–7708.

Kanojia, D., Garg, M., Gupta, S., Gupta, A., & Suri, A. (2009). Sperm-associated antigen 9, a novel biomarker for early detection of breast cancer. *Cancer Epidemiology and Prevention Biomarkers*, *18*(2), 630–639.

Keshavarzi, M., Darijani, M., Momeni, F., Moradi, P., Ebrahimnejad, H., Masoudifar, A., et al. (2017). Molecular imaging and oral cancer diagnosis and therapy. *Journal of Cellular Biochemistry*, *118*(10), 3055–3060.

Keshavarzi, M., Sorayayi, S., Jafar Rezaei, M., Mohammadi, M., Ghaderi, A., Rostamzadeh, A., et al. (2017). MicroRNAs-based imaging techniques in cancer diagnosis and therapy. *Journal of Cellular Biochemistry*, *29*(10), 26012.

Khalkhali, I., Mena, I., & Diggles, L. (1994). Review of imaging techniques for the diagnosis of breast cancer: A new role of prone scintimammography using technetium-99 mL sestamibi. *European Journal of Nuclear Medicine*, *21*(4), 357–362.

Khanmohammadi, R., Mir, F., Baniebrahimi, G., & Mirzaei, H. (2017). Oral tumors in children: Diagnosis and management. *Journal of Cellular Biochemistry*, *3*(10), 26316.

Kinkel, K., & Vlastos, G. (2001). MR imaging: Breast cancer staging and screening. *Seminars in Surgical Oncology*, *20*(3), 187–196.

Kioulafa, M., Kaklamanis, L., Stathopoulos, E., Mavroudis, D., Georgoulias, V., & Lianidou, E. (2009). Kallikrein 10 (KLK10) methylation as a novel prognostic biomarker in early breast cancer. *Annals of Oncology*, *20*(6), 1020–1025.

Kleer, C. G., Cao, Q., Varambally, S., Shen, R., Ota, I., Tomlins, S. A., et al. (2003). EZH2 is a marker of aggressive breast cancer and promotes neoplastic transformation of breast epithelial cells. *Proceedings of the National Academy of Sciences*, *100*(20), 11606–11611.

Kohler, C., Radpour, R., Barekati, Z., Asadollahi, R., Bitzer, J., Wight, E., et al. (2009). Levels of plasma circulating cell free nuclear and mitochondrial DNA as potential biomarkers for breast tumors. *Molecular Cancer*, *8*(1), 105.

Kristiansen, G., Winzer, K.-J., Mayordomo, E., Bellach, J., Schlüns, K., Denkert, C., et al. (2003). CD24 expression is a new prognostic marker in breast cancer. *Clinical Cancer Research*, *9*(13), 4906–4913.

Kumar, R., Zhuang, H., Schnall, M., Conant, E., Damia, S., Weinstein, S., et al. (2006). FDG PET positive lymph nodes are highly predictive of metastasis in breast cancer. *Nuclear Medicine Communications*, *27*(3), 231–236.

Kuroki-Suzuki, S., Kuroki, Y., Nasu, K., Nawano, S., Moriyama, N., & Okazaki, M. (2007). Detecting breast cancer with non-contrast MR imaging: Combining diffusion-weighted and STIR imaging. *Magnetic Resonance in Medical Sciences*, *6*(1), 21–27.

Kurozumi, S., Yamaguchi, Y., Kurosumi, M., Ohira, M., Matsumoto, H., & Horiguchi, J. (2017). Recent trends in microRNA research into breast cancer with particular focus on the associations between microRNAs and intrinsic subtypes. *Journal of Human Genetics*, *62*(1), 15–24.

Lüftner, D., & Possinger, K. (2002). Nuclear matrix proteins as biomarkers for breast cancer. *Expert Review of Molecular Diagnostics*, *2*(1), 23–31.

Lai, J. C. Y., Soh, C. B., Gunawan, E., & Low, K. S. (2011). UWB microwave imaging for breast cancer detection–experiments with heterogeneous breast phantoms. *Progress In Electromagnetics Research M*, *16*, 19–29.

Li, J., Zhao, J., Yu, X., Lange, J., Kuerer, H., Krishnamurthy, S., et al. (2005). Identification of biomarkers for breast cancer in nipple aspiration and ductal lavage fluid. *Clinical Cancer Research*, *11*(23), 8312–8320.

Li, Y., Wang, J. P., Santen, R. J., Kim, T. H., Park, H., Fan, P., et al. (2010). Estrogen stimulation of cell migration involves multiple signaling pathway interactions. *Endocrinology*, *151*(11), 5146–5156.

Malkas, L. H., Herbert, B. S., Abdel-Aziz, W., Dobrolecki, L. E., Liu, Y., Agarwal, B., et al. (2006). A cancer-associated PCNA expressed in breast cancer has implications as a potential biomarker. *Proceedings of the National Academy of Sciences*, *103*(51), 19472–19477.

Mann, R. M., Kuhl, C. K., Kinkel, K., & Boetes, C. (2008). Breast MRI: Guidelines from the european society of Breast imaging. *European Radiology*, *18*(7), 1307–1318. https://doi.org/10.1007/s00330-008-0863-7. Epub 2008 Apr 4.

Martínez-Galán, J., Torres, B., del Moral, R., Muñoz-Gámez, J. A., Martín-Oliva, D., Villalobos, M., et al. (2008). Quantitative detection of methylated ESR1 and 14-3-3-σ gene promoters in serum as candidate biomarkers for diagnosis of breast cancer and evaluation of treatment efficacy. *Cancer Biology & Therapy*, *7*(6), 958–965.

Mashreghi, M., Azarpara, H., Bazaz, M. R., Jafari, A., Masoudifar, A., Mirzaei, H., et al. (2017). Angiogenesis biomarkers and their targeting ligands as potential targets for tumor angiogenesis. *Journal of Cellular Physiology*, *13*(10), 26049.

Masoudi, M. S., Mehrabian, E., & Mirzaei, H. (2017). MiR-21: A key player in glioblastoma pathogenesis. *Journal of Cellular Biochemistry*, *20*(10), 26300.

Mathelin, C., Cromer, A., Wendling, C., Tomasetto, C., & Rio, M.-C. (2006). Serum biomarkers for detection of breast cancers: A prospective study. *Breast Cancer Research and Treatment*, *96*(1), 83–90.

Mattie, M. D., Benz, C. C., Bowers, J., Sensinger, K., Wong, L., Scott, G. K., et al. (2006). Optimized high-throughput microRNA expression profiling provides novel biomarker assessment of clinical prostate and breast cancer biopsies. *Molecular Cancer*, *5*(1), 24.

McBride, T. O. (2001). *Spectroscopic reconstructed near infrared tomographic imaging for breast cancer diagnosis: Dartmouth College*. Dartmouth College, New Hampshire: Thayer School of Engineering.

McDonnell, D. P., & Norris, J. D. (2002). Connections and regulation of the human estrogen receptor. *Science*, *296*(5573), 1642–1644.

Mehra, R., Varambally, S., Ding, L., Shen, R., Sabel, M. S., Ghosh, D., et al. (2005). Identification of GATA3 as a breast cancer prognostic marker by global gene expression meta-analysis. *Cancer Research*, *65*(24), 11259–11264.

Michell, M., Iqbal, A., Wasan, R., Evans, D., Peacock, C., Lawinski, C., et al. (2012). A comparison of the accuracy of film-screen mammography, full-field digital mammography, and digital breast tomosynthesis. *Clinical Radiology*, *67*(10), 976–981.

Mintun, M., Welch, M., Siegel, B., Mathias, C., Brodack, J., McGuire, A., et al. (1988). Breast cancer: PET imaging of estrogen receptors. *Radiology*, *169*(1), 45–48.

Mirzaei, H., Khataminfar, S., Mohammadparast, S., Sales, S. S., Maftouh, M., Mohammadi, M., et al. (2016). Circulating microRNAs as potential diagnostic biomarkers and therapeutic targets in gastric cancer: Current status and future perspectives. *Current Medicinal Chemistry*, *23*(36), 4135–4150.

Mirzaei, H., Fathullahzadeh, S., Khanmohammadi, R., Darijani, M., Momeni, F., Masoudifar, A., et al. (2017). State of the art in microRNA as diagnostic and therapeutic biomarkers in chronic lymphocytic leukemia. *Journal of Cellular Physiology*, *13*(10), 25799.

Mirzaei, H., Masoudifar, A., Sahebkar, A., Zare, N., Sadri Nahand, J., Rashidi, B., et al. (2017). MicroRNA: A novel target of curcumin in cancer therapy. *Journal of Cellular Physiology*, *15*(10), 26055.

Mirzaei, H., Ferns, G. A., Avan, A., & Mobarhan, M. G. (2017). Cytokines and MicroRNA in Coronary Artery Disease. *Advances in Clinical Chemistry*, *82*, 47–70.

Mirzaei, H., Momeni, F., Saadatpour, L., Sahebkar, A., Goodarzi, M., Masoudifar, A., et al. (2017). MicroRNA: Relevance to stroke diagnosis, prognosis, and therapy. *Journal of Cellular Physiology*, *9*(10), 25787.

Mirzaei, H., Sahebkar, A., Jaafari, M. R., Goodarzi, M., & Mirzaei, H. R. (2017). Diagnostic and therapeutic potential of exosomes in cancer: The beginning of a new tale? *Journal of Cellular Physiology*, *232*(12), 3251–3260.

Mirzaei, H. (2017). Stroke in women: Risk factors and clinical biomarkers. *Journal of Cellular Biochemistry*, *12*(10), 26130.

10974652, 2018, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jcp.26379 by Central University Of Rajasthan, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Mohammadi, M., Goodarzi, M., Jaafari, M. R., Mirzaei, H. R., & Mirzaei, H. (2016). Circulating microRNA: A new candidate for diagnostic biomarker in neuroblastoma. *Cancer Gene Therapy*, *23*(11), 371–372.

Moradian Tehrani, R., Verdi, J., Noureddini, M., Salehi, R., Salarinia, R., Mosalaei, M., et al. (2017). Mesenchymal stem cells: A new platform for targeting suicide genes in cancer. *Journal of Cellular Physiology*, *13*(10), 26094.

Moridikia, A., Mirzaei, H., Sahebkar, A., & Salimian, J. (2017). MicroRNAs: Potential candidates for diagnosis and treatment of colorectal cancer. *Journal of Cellular Physiology*, *16*(10), 25801.

Nagata, T., Shimada, Y., Sekine, S., Moriyama, M., Hashimoto, I., Matsui, K., et al. (2017). KLF4 and NANOG are prognostic biomarkers for triple-negative breast cancer. *Breast Cancer*, *24*(2), 326–335.

Orel, S. G., Mendonca, M. H., Reynolds, C., Schnall, M. D., Solin, L. J., & Sullivan, D. C. (1997). MR imaging of ductal carcinoma in situ. *Radiology*, *202*(2), 413–420.

Ozmen, N., Dapp, R., Zapf, M., Gemmeke, H., Ruiter, N. V., & van Dongen, K. W. (2015). Comparing different ultrasound imaging methods for breast cancer detection. *IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control*, *62*(4), 637–646.

Pan, Q., Bao, L. W., Kleer, C. G., Sabel, M. S., Griffith, K. A., Teknos, T. N., et al. (2005). Protein kinase Cε is a predictive biomarker of aggressive breast cancer and a validated target for RNA interference anticancer therapy. *Cancer Research*, *65*(18), 8366–8371.

Park, I. H., Kang, J. H., Lee, K. S., Nam, S., Ro, J., & Kim, J. H. (2014). Identification and clinical implications of circulating microRNAs for estrogen receptor-positive breast cancer. *Tumour Biology*, *35*(12), 12173–12180.

Pasanisi, P., Venturelli, E., Morelli, D., Fontana, L., Secreto, G., & Berrino, F. (2008). Serum insulin-like growth factor-I and platelet-derived growth factor as biomarkers of breast cancer prognosis. *Cancer Epidemiology and Prevention Biomarkers*, *17*(7), 1719–1722.

Pawlowski, V., Révillion, F., Hebbar, M., Hornez, L., & Peyrat, J.-P. (2000). Prognostic value of the type I growth factor receptors in a large series of human primary breast cancers quantified with a real-time reverse transcription-polymerase chain reaction assay. *Clinical Cancer Research*, *6*(11), 4217–4225.

Pisano, E. D., Gatsonis, C., Hendrick, E., Yaffe, M., Baum, J. K., Acharyya, S., et al. (2005). Diagnostic performance of digital versus film mammography for breast-cancer screening. *The New England Journal of Medicine*, *2005*(353), 1773–1783.

Pories, S. E., Zurakowski, D., Roy, R., Lamb, C. C., Raza, S., Exarhopoulos, A., et al. (2008). Urinary metalloproteinases: Noninvasive biomarkers for breast cancer risk assessment. *Cancer Epidemiology and Prevention Biomarkers*, *17*(5), 1034–1042.

Rabieian, R., Boshtam, M., Zareei, M., Kouhpayeh, S., Masoudifar, A., & Mirzaei, H. (2017). Plasminogen activator inhibitor type-1 as a regulator of fibrosis. *Journal of Cellular Biochemistry*, *18*(10), 26146.

Rashidi, B., Malekzadeh, M., Goodarzi, M., Masoudifar, A., & Mirzaei, H. (2017). Green tea and its anti-angiogenesis effects. *Biomedicine and Pharmacotherapy*, *89*, 949–956.

Ristimäki, A., Sivula, A., Lundin, J., Lundin, M., Salminen, T., Haglund, C., et al. (2002). Prognostic significance of elevated cyclooxygenase-2 expression in breast cancer. *Cancer Research*, *62*(3), 632–635.

Rui, Z., Jian-Guo, J., Yuan-Peng, T., Hai, P., & Bing-Gen, R. (2003). Use of serological proteomic methods to find biomarkers associated with breast cancer. *Proteomics*, *3*(4), 433–439.

Saadatpour, Z., Bjorklund, G., Chirumbolo, S., Alimohammadi, M., Ehsani, H., Ebrahiminejad, H., et al. (2016). Molecular imaging and cancer gene therapy. *Cancer Gene Therapy*, *18*(10), 62.

Saadatpour, L., Fadaee, E., Fadaei, S., Nassiri Mansour, R., Mohammadi, M., Mousavi, S. M., et al. (2016). Glioblastoma: Exosome and microRNA as novel diagnosis biomarkers. *Cancer Gene Therapy*, *23*(12), 415–418.

Saadatpour, Z., Rezaei, A., Ebrahimnejad, H., Baghaei, B., Bjorklund, G., Chartrand, M., et al. (2017). Imaging techniques: New avenues in cancer gene and cell therapy. *Cancer Gene Therapy*, *24*(1), 1–5.

Sadr-Nabavi, A., Ramser, J., Volkmann, J., Naehrig, J., Wiesmann, F., Betz, B., et al. (2009). Decreased expression of angiogenesis antagonist EFEMP1 in sporadic breast cancer is caused by aberrant promoter methylation and points to an impact of EFEMP1 as molecular biomarker. *International Journal of Cancer*, *124*(7), 1727–1735.

Saeedi Borujeni, M. J., Esfandiary, E., Taheripak, G., Codoner-Franch, P., Alonso-Iglesias, E., & Mirzaei, H. (2017). Molecular aspects of diabetes mellitus: Resistin, microRNA, and exosome. *Journal of Cellular Biochemistry*, *8*(10), 26271.

Saha Roy, S., & Vadlamudi, R. K. (2012). Role of estrogen receptor signaling in breast cancer metastasis. *International Journal of Breast Cancer*, *654698*(10), 19.

Santos, J. C., Ribeiro, M. L., Sarian, L. O., Ortega, M. M., & Derchain, S. F. (2016). Exosomes-mediate microRNAs transfer in breast cancer chemoresistance regulation. *American Journal of Cancer Research*, *6*(10), 2129–2139.

Sardanelli, F., Boetes, C., Borisch, B., Decker, T., Federico, M., Gilbert, F. J., et al. (2010). Magnetic resonance imaging of the breast: Recommendations from the EUSOMA working group. *European Journal of Cancer*, *46*(8), 1296–1316.

Scholzen, T., & Gerdes, J. (2000). The Ki-67 protein: From the known and the unknown. *Journal of Cellular Physiology*, *182*(3), 311–322.

Schrohl, A. S., Pedersen, H. C., Jensen, S. S., Nielsen, S. L., & Brunner, N. (2011). Human epidermal growth factor receptor 2 (HER2) immunoreactivity: Specificity of three pharmacodiagnostic antibodies. *Histopathology*, *59*(5), 975–983.

Schurpf, T., Chen, Q., Liu, J. H., Wang, R., Springer, T. A., & Wang, J. H. (2012). The RGD finger of Del-1 is a unique structural feature critical for integrin binding. *FASEB Journal*, *26*(8), 3412–3420.

Sempere, L. F., Keto, J., & Fabbri, M. (2017). Exosomal MicroRNAs in Breast cancer towards diagnostic and therapeutic applications. *Cancers*, *9*(7).

Shaker, O., Maher, M., Nassar, Y., Morcos, G., & Gad, Z. (2015). Role of microRNAs-29b-2, -155, -197 and -205 as diagnostic biomarkers in serum of breast cancer females. *Gene*, *560*(1), 77–82.

Shen, J., Hu, Q., Schrauder, M., Yan, L., Wang, D., Medico, L., et al. (2014). Circulating miR-148b and miR-133a as biomarkers for breast cancer detection. *Oncotarget*, *5*(14), 5284.

Shimono, Y., Zabala, M., Cho, R. W., Lobo, N., Dalerba, P., Qian, D., et al. (2009). Downregulation of miRNA-200c links breast cancer stem cells with normal stem cells. *Cell*, *138*(3), 592–603.

Simonian, M., Mosallayi, M., & Mirzaei, H. (2017). Circulating miR-21 as novel biomarker in gastric cancer: Diagnostic and prognostic biomarker. *Journal of Cancer Research*, [Epub ahead of print].

Singh, R., Pochampally, R., Watabe, K., Lu, Z., & Mo, Y. Y. (2014). Exosome-mediated transfer of miR-10b promotes cell invasion in breast cancer. *Molecular Cancer*, *13*(256), 1476–4598.

Singletary, S. E. (2003). Rating the risk factors for breast cancer. *Annals of Surgery*, *237*(4), 474–482.

Skliris, G. P., Munot, K., Bell, S. M., Carder, P. J., Lane, S., Horgan, K., et al. (2003). Reduced expression of oestrogen receptor beta in invasive breast cancer and its re-expression using DNA methyl transferase inhibitors in a cell line model. *Journal of Pathology*, *201*(2), 213–220.

Smith, A. L., Iwanaga, R., Drasin, D. J., Micalizzi, D. S., Vartuli, R. L., Tan, A. C., et al. (2012). The miR-106b-25 cluster targets Smad7, activates TGF-beta signaling, and induces EMT and tumor initiating cell characteristics downstream of Six1 in human breast cancer. *Oncogene*, *31*(50), 5162–5171.

Sochor, M., Basova, P., Pesta, M., Dusilkova, N., Bartos, J., Burda, P., et al. (2014). Oncogenic microRNAs: MiR-155, miR-19a, miR-181b, and miR-24 enable monitoring of early breast cancer in serum. *BMC Cancer*, *14*(1), 448.

10974652, 2018, 7, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/jcp.26379 by Central University Of Rajasthan, Wiley Online Library on [31/08/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Soderstrom, C. E., Harms, S. E., Copit, D. S., Evans, W. P., Savino, D. A., Krakos, P. A., et al. (1996). Three-dimensional RODEO breast MR imaging of lesions containing ductal carcinoma in situ. *Radiology*, *201*(2), 427–432.

Soliman, N. A., & Yussif, S. M. (2016a). Ki-67 as a prognostic marker according to breast cancer molecular subtype. *Cancer Biology & Medicine*, *13*(4), 496–504.

Soliman, N. A., & Yussif, S. M. (2016b). Ki-67 as a prognostic marker according to breast cancer molecular subtype. *Cancer Biology & Medicine*, *13*(4), 496.

Song, C., Zhang, L., Wang, J., Huang, Z., Li, X., Wu, M., et al. (2016). High expression of microRNA-183/182/96 cluster as a prognostic biomarker for breast cancer. *Scientific Reports*, *6*, 24502.

Speirs, V., Carder, P. J., Lane, S., Dodwell, D., Lansdown, M. R., & Hanby, A. M. (2004). Oestrogen receptor beta: What it means for patients with breast cancer. *Lancet Oncology*, *5*(3), 174–181.

Strom, A., Hartman, J., Foster, J. S., Kietz, S., Wimalasena, J., & Gustafsson, J. A. (2004). Estrogen receptor beta inhibits 17beta-estradiol-stimulated proliferation of the breast cancer cell line T47D. *Proceedings of the National Academy of Sciences of the United States of America*, *101*(6), 1566–1571.

Sun, J. C., Liang, X. T., Pan, K., Wang, H., Zhao, J. J., & Li, J. J., et al. (2010). High expression level of EDIL3 in HCC predicts poor prognosis of HCC patients. *World Journal of Gastroenterology*, *16*(36), 4611–4615.

Sunami, E., Vu, A. T., Nguyen, S. L., Giuliano, A. E., & Hoon, D. S. (2008). Quantification of LINE1 in circulating DNA as a molecular biomarker of breast cancer. *Annals of the New York Academy of Sciences.*, *1137*(1), 171–174.

Toth, B., Nieuwland, R., Liebhardt, S., Ditsch, N., Steinig, K., Stieber, P., et al. (2008). Circulating microparticles in breast cancer patients: A comparative analysis with established biomarkers. *Anticancer Research*, *28*(2A), 1107–1112.

Trecate, G., Vergnaghi, D., Manoukian, S., Bergonzi, S., Scaperrotta, G., Marchesini, M., et al. (2006). MRI in the early detection of breast cancer in women with high genetic risk. *Tumori*, *92*(6), 517.

Turner, B. C., Krajewski, S., Krajewska, M., Takayama, S., Gumbs, A. A., Carter, D., et al. (2001). BAG-1: A novel biomarker predicting long-term survival in early-stage breast cancer. *Journal of Clinical Oncology*, *19*(4), 992–1000.

Uematsu, T., Yuen, S., Yukisawa, S., Aramaki, T., Morimoto, N., Endo, M., et al. (2005). Comparison of FDG PET and SPECT for detection of bone metastases in breast cancer. *AJR American Journal of Roentgenology*, *184*(4), 1266–1273.

van Kruchten, M., de Vries, E. F., Arts, H. J., Jager, N. M., Bongaerts, A. H., Glaudemans, A. W., et al. (2015). Assessment of estrogen receptor expression in epithelial ovarian cancer patients using 16α-18F-fluoro-17β-estradiol PET/CT. *Journal of Nuclear Medicine.*, *56*(1), 50–55.

Viehweg, P., Lampe, D., Buchmann, J., & Heywang-Köbrunner, S. (2000). In situ and minimally invasive breast cancer: Morphologic and kinetic features on contrast-enhanced MR imaging. Magnetic Resonance Materials in Physics. *Biology and Medicine*, *11*(3), 129–137.

Wahl, R. L., Siegel, B. A., Coleman, R. E., & Gatsonis, C. G. (2004). Prospective multicenter study of axillary nodal staging by positron emission tomography in breast cancer: A report of the staging breast cancer with PET Study Group. *Journal of Clinical Oncology*, *22*(2), 277–285.

Wang, G., Qin, Y., Zhang, J., Zhao, J., Liang, Y., Zhang, Z., et al. (2014). Nipple discharge of CA15-3, CA125, CEA and TSGF as a new biomarker panel for breast cancer. *International Journal of Molecular Sciences*, *15*(6), 9546–9565.

Weigel, M. T., & Dowsett, M. (2010). Current and emerging biomarkers in breast cancer: Prognosis and prediction. *Endocrine-Related Cancer*, *17*(4), 10–0136.

Wellings, E., Vassiliades, L., & Abdalla, R. (2016). Breast cancer screening for high-Risk patients of different ages and risk—Which modality is most effective? *Cureus*, *8*(12), 945.

Westerhof, J. P., Fischer, U., Moritz, J. D., & Oestmann, J. W. (1998). MR imaging of mammographically detected clustered microcalcifications: Is there any value? *Radiology*, *207*(3), 675–681.

Yamashita, H., Nishio, M., Toyama, T., Sugiura, H., Zhang, Z., Kobayashi, S., et al. (2003). Coexistence of HER2 over-expression and p53 protein accumulation is a strong prognostic molecular marker in breast cancer. *Breast Cancer Research*, *6*(1), R24.

Yarosh, W., Barrientos, T., Esmailpour, T., Lin, L., Carpenter, P. M., Osann, K., et al. (2008). TBX3 is overexpressed in breast cancer and represses p14ARF by interacting with histone deacetylases. *Cancer Research*, *68*(3), 693–699.

Ye, Y., Xiao, Y., Wang, W., Yearsley, K., Gao, J. X., & Barsky, S. H. (2008). ERalpha suppresses slug expression directly by transcriptional repression. *Biochemical Journal*, *416*(2), 179–187.

Yi, J. K., Chang, J. W., Han, W., Lee, J. W., Ko, E., Kim, D. H., et al. (2009). Autoantibody to tumor antigen, alpha 2-HS glycoprotein: A novel biomarker of breast cancer screening and diagnosis. *Cancer Epidemiology and Prevention Biomarkers*, *18*(5), 1357–1364.

Yu, F., Jiao, Y., Zhu, Y., Wang, Y., Zhu, J., Cui, X., et al. (2012). MicroRNA 34c gene down-regulation via DNA methylation promotes self-renewal and epithelial-mesenchymal transition in breast tumor-initiating cells. *Journal of Biological Chemistry*, *287*(1), 465–473.

Zanello, P. A., Robim, A. F. C., Oliveira, T. M. G. D., Elias Junior, J., Andrade, J. M. D., Monteiro, C. R., et al. (2011). Breast ultrasound diagnostic performance and outcomes for mass lesions using Breast Imaging Reporting and Data System category 0 mammogram. *Clinics*, *66*(3), 443–448.

Zhang, X., Wan, G., Mlotshwa, S., Vance, V., Berger, F. G., Chen, H., et al. (2010). Oncogenic Wip1 phosphatase is inhibited by miR-16 in the DNA damage signaling pathway. *Cancer Research*, *70*(18), 7176–7186.

Zhang, L., Song, X., Wang, X., Xie, Y., Wang, Z., Xu, Y., et al. (2015). Circulating DNA of HOTAIR in serum is a novel biomarker for breast cancer. *Breast Cancer Research and Treatment*, *152*(1), 199–208.

Zhang, G., Zhang, W., Li, B., Stringer-Reasor, E., Chu, C., Sun, L., et al. (2017). MicroRNA-200c and microRNA-141 are regulated by a FOXP3-KAT2B axis and associated with tumor metastasis in breast cancer. *Breast Cancer Research*, *19*(1), 017–0858.

Zhao, H., Shen, J., Medico, L., Wang, D., Ambrosone, C. B., Liu, S. (2010). A pilot study of circulating miRNAs as potential biomarkers of early stage breast cancer. *PLoS ONE*, *5*(10), e13735.

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